ARBITER

Where we stand, what happens next, and why this is not another prediction tool

Pfizer Digital & Technology Hackathon 2026 · Problem Statement 3 · Team BU 1 — Jack He, Andres Lopez, Jose Cruz-Lopez

Prepared 6 August 2026 · Submission due 16 August 2026

> Written for a reader who does not work in toxicology or software. Every number in it was measured from the running system on 6 August 2026, not estimated. Where the honest answer is unflattering, this document gives the honest answer — that is deliberate, and section 3 explains why it is the stronger position.

1. What ARBITER does, in plain English

Before a drug is given to a person, it is tested for liver damage in several different ways — a computer model that guesses from the molecule's shape, a dish of human liver cells, a test of whether the molecule blocks the liver's waste-export pump, and animal studies.

These tests routinely disagree. One says dangerous, another says fine. A scientist then has to decide which to believe. Today that reasoning happens in someone's head and in a meeting, and six months later nobody can reconstruct exactly why the call went the way it did.

ARBITER is the layer that does that reconciliation in the open. Given the conflicting evidence, it:

- reaches a **position** — advance, do not advance, or *not enough evidence to say*;
- shows **the argument** that led there, in steps a scientist can follow and dispute;
- names **the single piece of evidence that would change its mind**, and which experiment would produce it;
- records **who signed off**, in a log that cannot be quietly edited afterwards.

The rules are locked before the results are seen

ARBITER's reasoning follows six written rules — for example, *evidence from human cells outweighs evidence from animals when the question is about humans*, and *a test that measures the actual biological mechanism outweighs one that only notices the molecule looks suspicious*.

Those six rules were written down and sealed with a digital fingerprint **before** any results were measured. The system refuses to run if that fingerprint changes. This is the scientific equivalent of registering your predictions before the experiment — it makes it impossible to quietly adjust the rules afterwards to make the answer look better, and it means nobody has to take our word for that.

2. The differentiator, stated plainly

> Everyone else at this hackathon will build something that **predicts** toxicity.
> ARBITER reasons through the conflicts **between** those predictions. It is not another predictor. It is the layer that adjudicates — and the human still signs.

DIMENSION	A TYPICAL PREDICTION TOOL	ARBITER
What it produces	A score, or a red/green flag	A position, the argument behind it, and what would overturn it
When evidence conflicts	Averages it away, or picks a favourite input	Adjudicates it explicitly, by rules you can read and argue with
When evidence is thin	Still answers, with a confident-looking number	Says *not enough evidence* — and says exactly what is missing
Can the goalposts move?	Yes — thresholds are usually tuned until results look good	No — the rules are fingerprinted and the system refuses to run if they change
Ask it twice	An AI-based tool may answer differently each time	Identical answer, every time, provably
Can a scientist argue back?	Rarely — the model is fixed and opaque	Yes — challenge a rule in plain English and watch the conclusion change live
Afterwards	A number in a spreadsheet	A signed, tamper-evident record of who decided what, and on what basis

The three that are hardest for anyone else to copy

1. **It is willing to say "I don't know."** Almost no tool does this, because abstaining looks like failure in a demo. It is the opposite: a system that declines when the evidence is genuinely insufficient is one a scientist can actually trust when it *does* commit.
2. **It tells you which experiment to run next — and why.** Not "which test is usually informative", but *"this specific rule is what is blocking a conclusion, and here is the evidence that would overturn that specific rule."* This is the genuinely uncommon mechanism in the system.
3. **The rules were sealed before the results.** Anyone can claim their model wasn't tuned to flatter itself. We can prove it.

3. Where it honestly stands

This section exists because the temptation to soften it should be resisted, and because a judge who discovers a softened claim discounts everything said afterwards.

> **ARBITER does not beat the best comparison. It ties one, exactly.**
>
> Measured on 267 compounds it had never seen: ARBITER matches a simple baseline on every single measure. **And the reason is measurable, which makes it a better story than the bare fact.** Both are scoring the *same four compounds* — there are only four compounds in the whole set carrying the one kind of evidence strong enough to decide on. An exact tie between two methods judged on an identical set of four is close to expected, not a coincidence.

ARBITER declines to commit on 97.4% of compounds. That sounds like failure and is in fact the finding. The reason is not that the reasoning is broken — it is that **the evidence base is too thin to license a decision**. Half the compounds carry exactly *one* piece of evidence. ARBITER exists to adjudicate between sources; 140 of the 267 have only one source. It is being asked to do its job where its job does not exist.

The sharpest version: for **254 of the 260 declines**, even if every piece of evidence were restated at maximum confidence, it still could not reach the bar. The system declines *before it reads a single value*. That is a structural fact about the data, and it is provable.

> **The number worth leading with**
>
> **Shake every expert judgement in the system by up to 50% in either direction, and the recommended next experiment stays the same 99.2% of the time.** The advice does not depend on getting the numbers exactly right — it depends on the structure of the argument. That is a robustness claim very few tools can make.

4. What happens next

Ten days remain. Work splits into three groups by what is blocking it.

Group A — unblocked, costs nothing, can start today

ITEM	WHAT IT DOES, PLAINLY	STATUS
Automate the data safety-check	A set of checks proves we never let the system peek at its own exam answers — the single claim every reported number depends on. Today those checks only run when someone remembers to run them by hand.	Small job, nothing blocking
Build the AI-comparison scaffolding	The scoring, testing and prompt work for the "why not just ask an AI?" experiment. Needs no AI account and no money — and is most of the work.	Fully specified, not built

Group B — needs a decision from you

ITEM	WHAT IT DOES, PLAINLY	BLOCKED BY
Run the AI comparison	Ask a strong AI the same question 25 times per compound and measure how often it contradicts itself. Turns our central argument into a measured fact.	An AI account and roughly $27
Real-world dose data	Knowing the dose a patient actually receives lets the system tell "tested at a realistic dose" from "tested at a dose nobody would ever take". This is the single biggest cause of the 97.4% decline rate.	Data we do not have; deadline passed

Group C — needs a person, not an engineer

- **Build the submission package from a clean copy on a different machine** and click through it by hand. A previous version shipped as a blank page; this check is why it will not happen twice.
- **Build the deck from the real measured numbers**, pulled directly from the system rather than retyped.
- **Record a walkthrough** as insurance against a live demo failing in the room.
- **Rehearse, and check it is readable** when shared on a call.
- **Submit early.**

5. Where AI could genuinely differentiate this

The rule that must not be broken

AI does language, never judgment. Reading a sentence and matching a question to a place on screen are language tasks. Deciding whether a drug is safe is not. Every idea below keeps the AI on the language side of that line. The moment an AI makes the call, the entire pre-registration argument — the thing that makes ARBITER defensible — collapses. This is not caution; it *is* the differentiator.

1. Turn the AI comparison into the centrepiece, not a footnote

SPECIFIED

WHAT IT IS

Ask a top-tier AI the same question, with the same evidence and the same rules, 25 times. Show on screen that it returns different verdicts to identical input. Then show ARBITER returning one answer, every time.

WHY IT WINS

Every other entry will say "we used AI." This says *"we measured what AI cannot do here, and built the thing that fixes it."* It converts the most likely hostile question in the room — *why not just ask ChatGPT?* — into the strongest moment of the demo. Right now that question has only an argument as an answer, not a measurement.

COST Roughly $27 and a day of engineering.

2. An AI devil's advocate that cannot make things up

NEW

WHAT IT IS

A button that asks an AI to build the strongest possible case *against* ARBITER's position — but restricted to citing evidence that already exists in the file. It is not allowed to write free text, only to point at real claims. The engine then evaluates whether that counter-case actually defeats anything.

WHY IT WINS

It is a system that actively tries to prove itself wrong, in front of the judges, and shows the result either way. It also demonstrates the anti-fabrication design in the most vivid possible setting: the AI is *structurally unable* to invent evidence, because the only thing it is permitted to return is a list of identifiers. AI stress-tests; the engine still judges.

COST Moderate — it reuses machinery that already exists.

3. AI that reads the literature and proposes structured evidence NEW

WHAT IT IS

Paste a paper's abstract; an AI proposes a structured evidence entry — what was measured, in what system, at what dose. A human approves or rejects it before it counts. The system already **refuses to accept** any entry claiming real-world dose relevance without a cited source, so the AI cannot smuggle in an unsupported claim even if it wanted to.

WHY IT WINS

This is the only idea here that attacks **the actual measured weakness**. The 97.4% decline rate is caused by thin evidence, not bad reasoning. An AI that widens the evidence base is using AI to fix the real problem rather than to decorate the demo — and it turns section 3's uncomfortable finding into a roadmap.

COST Highest of the three. Probably a post-submission story unless scoped very tightly.

4. "Argue with it in plain English" — already built, undersold BUILT

WHAT IT IS

Type "*that mouse study shouldn't count for a human question*" and the system proposes a specific change, **shows you the change before applying it**, and recomputes the conclusion live when you accept.

WHY IT WINS

Contestability. Most AI tools ask you to accept their output; this one lets a scientist overrule it in their own words and records the override. Critically, the AI never touches the reasoning — it only translates English into a proposed change. If the AI is unavailable, four progressively simpler fallbacks handle it and the reasoning is identical. **The demo cannot fail because the network did.**

COST Already working. Needs framing in the deck, not engineering.

5. "Could you have caught it at the time?" SPECIFIED

WHAT IT IS

Take a drug that was later found to cause liver injury. Feed the system *only* the evidence that existed before that was known, and show what it would have said at the time.

WHY IT WINS

It is the most intuitive possible demonstration for a non-technical judge, and it is a genuine methodological contribution — testing an evidence-integration tool on a historical decision point is a validation design, not just a demo trick. AI plays little part here; the story is the award-shaped part.

COST Low to moderate. The replay machinery exists; picking a compelling compound is the work.

If you only do one

Idea 1. It is already fully specified, costs about $27, directly answers the question most likely to be asked in the room, and is the only one that converts our central claim from an argument into a measurement. Ideas 2 and 5 are the best value after that; idea 3 is the strongest *next-year* story and is worth naming aloud as the roadmap.

Three things that would cost more credibility than they gain

Letting an AI make the call — it destroys the pre-registration argument and the determinism claim in one move. **Adjusting the decision threshold** to reduce the 97.4% decline rate — it was sealed precisely so it could not be moved after seeing that number, and moving it substantially buys only six compounds. **Claiming a first** — the individual components are all precedented, and a judge who knows the field will discount everything said afterwards. What is new is the assembly, and that is defensible.
